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## „ESG Scores and Financial Performance of Banking and Energy firms“

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# Abstract

This thesis investigates whether ESG scores are systematically related to financial performance in European banking and energy firms. Drawing on asset pricing theory and multi-factor models, it examines whether ESG characteristics alter the traditional risk–return relationship or proxy for systematic risk and investor preferences.

The empirical analysis is based on 43 firms from the STOXX Europe 600 Banks and STOXX Europe 600 Oil & Gas indices over 2010–2024. ESG performance is measured using Bloomberg and LSEG scores, while financial outcomes include valuation ratios, profitability measures, stock returns, and volatility. Pooled OLS and fixed effects panel regressions are employed to distinguish cross-sectional differences from within-firm and market-wide effects.

Results show that ESG effects are often significant in pooled OLS but weaken substantially under firm and time fixed effects, indicating that cross-sectional differences rather than causal ESG improvements drive much of the association. Higher ESG scores are linked to lower volatility, particularly for banks, but frequently to lower market valuation. No robust evidence of a systematic ESG return premium is found, and results differ across sectors and ESG providers.

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# 1. Introduction

> "As a responsible asset manager, it is our duty to constantly take the pulse of ESG investing dynamics, to ensure that we remain at the forefront of responsible investing. Indeed, these findings show that investing with an ESG lens is a complex but ultimately rewarding process."
>
> (Amundi S.A., 2019a)

Environmental, Social, and Governance (ESG) considerations have become an increasingly prominent component of financial analysis and investment decision-making. Statements such as this, made by Vincent Mortier, Group CIO of Amundi Asset Management, have become more frequent in recent years and contribute to the perception that ESG integration has become indispensable in modern finance. Investors, regulators, and financial institutions alike now place growing emphasis on non-financial factors, particularly in sectors characterized by high regulatory intensity and environmental exposure, such as banking and energy. Consequently, ESG scores are widely employed as indicators of firms' sustainability performance and perceived risk profiles (Amundi S.A., 2019b).

At the same time, the increasing prominence of ESG considerations raises the question of whether their relevance in financial markets is as substantial as often suggested. In practice, ESG integration is frequently emphasized by asset managers as a component of risk assessment and long-term resilience rather than as a direct driver of superior short-term returns. While higher ESG standards may improve firms' governance structures or reduce exposure to certain non-financial risks, they may also entail additional costs, regulatory constraints, or operational trade-offs that could adversely affect financial performance (Amundi S.A., 2025).

Against this background, the question of whether higher ESG scores systematically translate into superior—or at least not inferior—financial performance remains an open empirical issue. While ESG integration is often associated with improved risk management and long-term resilience, critics argue that stronger ESG engagement may constrain firms through higher compliance costs, restricted investment universes, or foregone profitable activities.

This debate gives rise to the central research question of this thesis: Do higher ESG scores inevitably lead to lower financial returns?

To address this question, the thesis examines the relationship between ESG scores and financial performance in European firms operating in the banking and energy sectors. By focusing on industries characterized by distinct regulatory environments and economic structures, the analysis provides a sector-specific perspective on the ESG–performance relationship. Overall, the objective is to empirically assess whether higher ESG scores are associated with systematic differences in firms' financial performance and, in particular, whether such differences imply a return trade-off or not, without presuming the direction of the effect ex ante.

# 2. Literature review

The following chapter reviews the existing literature on environmental risk and risk factor models, drawing on the foundational work of Harry Markowitz and the subsequent formalization of asset pricing theory by William F. Sharpe. In the academic discourse, this line of research has attracted growing attention, giving rise to a variety of empirical approaches and theoretical extensions. Despite this increasing interest, it remains unresolved whether environmental risk—across its various definitions and measurement approaches—can be systematically incorporated into established asset pricing frameworks, such as the factor models proposed by Eugene F. Fama and Kenneth R. French or the momentum-augmented model developed by Mark Carhart. (Fama and French, 1992, 2015)

In light of the objectives of this thesis, which seeks to examine the effect of sustainability ratings as a potential risk factor for stock returns, the following subchapters serve two purposes. First, they provide a historical overview of the development of established risk factors in asset pricing theory. Second, they outline the evolution of sustainability-related concepts and rating approaches, thereby setting the context for the subsequent empirical analysis. It should be acknowledged that the studies discussed in this chapter do not constitute an exhaustive review of the literature. Rather, the selection reflects the author's judgment in highlighting representative contributions within a broad and specialized field of academic research.

## 2.1. Risk and return in theory

"The expected return on a stock contains compensation for bearing the risk of the return." (Fama, 2013)

The Capital Asset Pricing Model (CAPM), introduced by William F. Sharpe in Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk, represents one of the foundational frameworks for analyzing the relationship between risk and expected return. Building on earlier portfolio theory, the model formalizes the idea that not all sources of risk are equally relevant for asset pricing. While firm-specific, idiosyncratic risks can be mitigated through diversification, only risk components related to overall market fluctuations are systematically priced. (Sharpe, 1964)

Within this framework, the expected return of an asset is determined by its sensitivity to market-wide movements, captured by the beta coefficient. Beta measures the degree to which an asset's return co-moves with the market portfolio and thus reflects its exposure to non-diversifiable risk. Assets with higher market sensitivity are therefore expected to offer higher returns as compensation for bearing greater systematic risk. This conceptualization of risk as a market-related phenomenon has had a lasting influence on both academic research and practical investment analysis, serving as a cornerstone for subsequent developments in asset pricing theory. (Sharpe, 1964)

While the CAPM provides a clear and intuitive framework for linking expected returns to market risk, later empirical studies have shown that differences in returns across firms cannot be fully explained by market beta alone. In particular, systematic patterns in average stock returns were observed that remained unexplained within a single - factor setting. These findings motivated researchers to investigate whether additional sources of systematic variation could help explain differences in expected returns across assets. (Fama and French, 1992)

In this context, Eugene F. Fama and Kenneth R. French made an important contribution in their study The Cross-Section of Expected Stock Returns. They showed that firm size and the ratio of book equity to market equity are strongly related to average stock returns, even after controlling for market risk. Notably, stocks of smaller firms and firms with higher book-to-market ratios were found to earn higher average returns than predicted by the CAPM. (Fama and French, 1992)

To capture these regularities, Fama and French proposed a three-factor model that extends the market factor by including factors related to size and value. Shortly thereafter, Mark Carhart further extended this framework in On Persistence in Mutual Fund Performance by adding a momentum factor. This additional factor reflects the tendency of stocks with strong past performance to continue outperforming in the short to medium term. The inclusion of momentum improved the model's ability to explain return patterns, particularly those related to performance persistence. (Carhart, 1997)

Building on these developments, Fama and French later expanded their framework by introducing a five-factor model. In addition to market, size, and value, this model incorporates factors related to firms' profitability and investment behaviour. Empirical evidence suggests that firms with higher profitability and more conservative investment policies exhibit systematically different return patterns, further enhancing the explanatory power of the model. (Fama and French, 2015)

Importantly, Fama and French did not argue that firm characteristics such as size or book-to-market ratios directly generate risk premia. Instead, these characteristics are interpreted as proxies for underlying sources of systematic risk, as reflected in their strong co-movement with asset returns. In this way, multi-factor models preserve the central insight of the CAPM—namely that expected returns compensate investors for bearing non-diversifiable risk—while allowing for a broader and more flexible representation of how such risk manifests in financial markets. (Fama and French, 2015)

## 2.2. ESG considerations and financial performance

ESG investing has emerged in parallel with a broader societal focus on sustainable development and long-term environmental and social challenges. International initiatives, such as the definition of the Sustainable Development Goals by the United Nations in 2015 and the increasing emphasis on climate-related risks highlighted in reports by scientific bodies, have contributed to a growing awareness of sustainability issues across economic and political systems. Within this context, governments, corporations, and individuals are commonly viewed as key actors in shaping a more sustainable development path. (Bennani et al., 2018)

From a financial perspective, this shift has also placed asset owners and asset managers under increasing scrutiny. Responsible investment frameworks, most notably the United Nations-supported Principles for Responsible Investment (PRI), reflect the growing role of institutional investors in promoting the integration of environmental, social, and governance considerations into investment decision-making. Since its establishment, the PRI initiative has attracted a steadily expanding network of signatories, indicating the rising relevance of ESG considerations within global capital markets. (Bennani et al., 2018)

Following the global financial crisis, the concept of green finance gained further prominence and began to influence the structure of financial markets more broadly. While green finance typically refers to the funding of environmentally oriented projects and assets, ESG investing adopts a more comprehensive perspective by incorporating not only environmental aspects but also social and governance dimensions. As a result, ESG investing is increasingly viewed as a holistic framework for evaluating corporate behaviour and long-term sustainability. (Bennani et al., 2018)

Motivations for ESG investing are generally twofold. On the one hand, investors may be driven by ethical considerations, such as promoting social inclusion, environmental protection, or responsible corporate conduct. On the other hand, ESG integration is often justified as a tool for managing long-term risks, including regulatory, reputational, operational, and financial risks. These motivations are closely intertwined, as ethical concerns and risk considerations frequently overlap in practice. (Bennani et al., 2018)

Despite the growing adoption of ESG strategies, the implications of ESG investing for financial performance remain subject to debate. While some market participants argue that performance considerations are secondary to broader sustainability objectives, institutional investors are bound by fiduciary duties that encompass both financial and non-financial responsibilities. As a result, understanding how ESG integration influences investment performance remains an important topic within academic research and professional asset management. (Bennani et al., 2018)

In this context, Gantchev et al. provided empirical evidence on the potential trade-off between sustainability considerations and financial performance by examining the introduction of Morningstar's sustainability ratings for mutual funds. These ratings summarized the ESG characteristics of a fund's underlying holdings and assigned funds to discrete rating categories that were easily observable by investors. Gantchev et al. showed that, following the introduction of these ratings, fund managers—particularly those close to rating thresholds—adjusted their portfolios toward stocks with higher ESG scores in an attempt to improve their sustainability ratings and attract investor inflows. (Gantchev et al., 2024)

In the short term, these adjustments influenced fund flows, indicating that investors initially responded to sustainability ratings when allocating capital. However, the portfolio reallocations required to achieve higher sustainability scores were often associated with weaker subsequent performance, as highly rated ESG stocks tended to experience increased demand and lower future returns. Over the longer term, investors placed greater weight on realized fund performance rather than sustainability ratings, leading to a decline in the importance of sustainability scores for fund flows. Overall, the findings suggested that while sustainability ratings affected investment behaviour in the short run, financial performance considerations dominated in the longer run, thereby limiting fund managers' incentives to prioritize sustainability ratings at the expense of returns. (Gantchev et al., 2024)

In a related line of research, Bennani et al. and a later study by Drei et al. examined the development of ESG investing between 2010 and 2019. Their results suggested that the relationship between ESG integration and portfolio performance changed over time. While ESG-focused strategies were linked to lower returns in both the US and European markets before 2013, this pattern shifted after 2014, when ESG assets began to contribute positively to returns. Overall, these findings reflected the broader, yet still mixed, evidence in the literature regarding the impact of ESG factors on investment performance (Bennani et al., 2018; Drei et al., 2019)

Building on this mixed evidence, Pedersen et al. provided a structured framework to analyse how ESG considerations can be integrated into portfolio choice and asset pricing. Their study examined ESG investing from the perspective of both financial trade-offs and investor preferences. Rather than treating ESG as separate from traditional finance, the authors showed how sustainability considerations can be incorporated into standard risk–return analysis. (Pedersen et al., 2021)

The paper introduced the concept of an ESG-efficient frontier, which extends the traditional efficient frontier by including ESG performance as an additional dimension. This framework illustrated that investors face a trade-off between expected returns, risk, and sustainability outcomes. For a given level of ESG performance, the ESG-efficient frontier identified portfolios that achieved the highest possible risk-adjusted returns. Importantly, this approach did not assume that higher ESG scores automatically lead to higher or lower returns. (Pedersen et al., 2021)

In addition, the authors developed an asset pricing framework in which ESG characteristics influenced expected returns depending on market conditions and investor preferences. If investors were willing to accept lower financial returns in exchange for higher ESG performance, assets with stronger ESG profiles could exhibit lower expected returns. At the same time, ESG scores were shown to capture information about firms' fundamentals and risk exposures, suggesting that ESG characteristics can be financially relevant. (Pedersen et al., 2021)

Consistent with earlier findings, Pástor et al. examined why environmentally sustainable assets delivered strong realized returns in recent years, even though theory predicts lower expected returns for such assets. The study showed that this apparent paradox was driven by shifts in environmental concerns that affected investor demand and discount rates, rather than by improvements in firms' fundamentals. (Pástor et al., 2022)

The authors documented this pattern across different asset classes. In the bond market, German green bonds outperformed comparable non-green bonds as the so-called greenium widened, reflecting investors' increased willingness to accept lower yields for environmentally sustainable securities. In equity markets, U.S. green stocks outperformed brown stocks during periods in which climate concerns intensified. These results suggested that environmental characteristics had a greater impact on asset prices when climate-related risks became more salient to investors. (Pástor et al., 2022)

Despite this strong realized performance, the study estimated lower expected returns for green assets relative to brown assets, aligning with standard asset pricing theory and highlighted the distinction between realized returns and expected returns. Green firms were shown to be less exposed to adverse environmental and regulatory shocks, which reduced their risk and consequently their anticipated reward. At the same time, increased investor demand raised prices of green assets, mechanically lowering their expected future returns. (Pástor et al., 2022)

## 2.3. ESG hedging and investor sentiment

Climate risk has become an increasingly important consideration for institutional investors and is no longer viewed solely as an ethical or long-term issue. Krueger et al. examined how institutional investors perceive climate-related risks and how these perceptions influence investment decisions. The study was based on a global survey of 439 institutional investors, including pension funds, insurance companies, asset managers, and sovereign wealth funds. (Krueger et al., 2020)

The survey results indicated that a large share of investors regarded climate risk as financially relevant. While traditional financial risks remained more prominent, climate-related risks were widely expected to affect asset values and investment outcomes, particularly through regulatory and policy developments. Importantly, many respondents did not consider climate risk a distant concern but believed that its effects had already begun to materialize or would do so in the near future. This perception suggested that climate risk influenced expectations about future market developments and downside risks. (Krueger et al., 2020)

Despite the growing empirical evidence that climate considerations should be taken into account, the study also highlighted the practical difficulties associated with integrating climate risks into the investment process. Many market participants, including institutional investors, reported that climate risks were difficult to price and hedge. These challenges were attributed to several factors, including the systematic nature of climate risks, limited and inconsistent disclosure by portfolio firms, and the lack of established investment tools and suitable hedging instruments. As a result, translating climate risk awareness into concrete portfolio strategies remained a complex task. (Krueger et al., 2020)

In response, investors indicated that they had begun to address climate risk through initial measures, motivated by a combination of considerations rather than a single dominant factor. These motivations included reputational concerns, regulatory and legal responsibilities, and expectations that climate-related developments could influence portfolio risk and returns. This mix of drivers suggested that climate risk was incorporated into investment processes for both financial and non-financial reasons. (Krueger et al., 2020)

In this setting, Engle et al. explored a market-based approach to climate-related risks. Although climate risk is increasingly perceived as financially relevant, investors face substantial difficulties in hedging this risk using conventional instruments. The authors argued that climate change represents a risk that is difficult to hedge directly because its economic effects are uncertain, long-term, and not tied to a single observable outcome. As a result, standard hedging approaches that rely on clearly defined risk realizations are often ineffective. (Krueger et al., 2020; Engle et al., 2020)

To address this issue, Engle et al. developed a framework that focuses on innovations in climate-related news rather than on distant climate outcomes. Using textual analysis of major newspapers, the authors constructed climate news indices that capture unexpected changes in climate-related concerns. They showed that asset prices respond systematically to these news shocks, indicating that climate risk is reflected in financial markets through changing investor expectations and discount rates. (Engle et al., 2020)

Building on this insight, the study demonstrated that firms differ in their exposure to climate news. In particular, companies with stronger environmental characteristics, proxied by environmental ESG scores, were found to be less negatively affected by adverse climate-related news. Based on these differences, Engle et al. constructed hedging portfolios that take long positions in firms with lower exposure to climate-related news and short positions in firms with higher exposure. These portfolios provided protection against negative climate news both in sample and out of sample. (Engle et al., 2020)

Importantly, the authors also showed that securities offering protection against climate risk tend to exhibit lower expected returns. This finding indicated that investors are willing to accept lower expected returns in exchange for hedging climate-related risks, consistent with standard asset pricing intuition. Overall, the analysis connected climate-related investor sentiment with ESG characteristics and hedging activity in financial markets. (Engle et al., 2020)

# 3. Empirical outlook

Building on the theoretical considerations discussed above, this study expects firms with weaker overall ESG scores to command higher expected returns over longer horizons. Lower ESG performance is associated with higher exposure to sustainability-related risks, including regulatory, operational, and reputational challenges, for which investors may require compensation in the form of a risk premium. Accordingly, firms with poorer ESG characteristics are expected to exhibit higher average returns relative to firms with stronger ESG performance.

At the same time, this expectation does not rule out periods in which firms with superior ESG profiles outperform their less sustainable counterparts. During phases of heightened attention to sustainability issues or shifts in investor sentiment, demand for ESG-aligned assets may increase, leading to temporary price effects and relative outperformance. Such episodes are likely to reflect changes in investor preferences rather than persistent differences in fundamental risk.

The empirical analysis that follows examines these expectations by assessing whether ESG scores are systematically associated with differences in stock performance. By comparing return patterns across firms with varying ESG characteristics, the study aims to evaluate whether sustainability-related risks are reflected in financial markets and how they influence return dynamics over time.

# 4. Data and Sample Construction

The empirical analysis is based on a sample of European publicly listed firms drawn from two sector-specific indices: the STOXX Europe 600 Banks and the STOXX Europe 600 Oil & Gas indices. These indices represent major firms operating in the European banking sector and the energy sector, respectively, and are widely used benchmarks for sector-level analysis within European equity markets.

The observation period spans from December 30, 2010 to December 30, 2024, allowing for the examination of ESG-related effects over a long horizon that includes different market conditions and regulatory environments. For the purpose of this study, the index composition at a fixed point in time was used throughout the entire sample period. While the constituents of both indices changed slightly over time, this approach ensures a consistent firm sample and avoids distortions arising from index rebalancing. The use of established sector indices reduces potential liquidity-related issues, as the firms included are typically large and regularly traded.

The focus on banking and oil & gas firms is motivated by their differing exposure to sustainability-related risks. Banks operate in a highly regulated environment and play a central role in capital allocation, while oil & gas firms are particularly exposed to environmental and transition-related challenges. This sectoral distinction allows for a comparative perspective on the relationship between ESG characteristics and stock performance.

All firm-level financial and market data used in the regression analysis, including stock returns and the model's control variables, were extracted from Bloomberg. ESG performance was measured using two separate data sources: the Bloomberg ESG Score and the ESG score provided by LSEG, accessed via Datastream. By incorporating ESG scores from two different providers, the analysis aims to mitigate potential data biases and reduce sensitivity to provider-specific methodologies, with the expectation that results based on multiple scoring systems are more resilient and less affected by methodological changes.

Overall, the resulting dataset consists of a panel of 43 European banking and energy firms observed over the period from 2010 to 2024 and forms the basis for the subsequent empirical investigation of ESG-related return patterns.

# 5. ESG Scoring Methodologies

This section outlines the general construction of the ESG scores used in the empirical analysis. The study relies on ESG ratings provided by Bloomberg and LSEG, which represent two widely used but methodologically distinct approaches to measuring corporate sustainability performance. Both providers aggregate a broad set of environmental, social, and governance indicators into composite scores, drawing on publicly available company disclosures, regulatory filings, and additional third-party information. While the overall objective of both scoring systems is to assess firms' ESG performance, they differ in terms of choice of underlying metrics, aggregation approaches, update frequency, etc. As a result, ESG scores from different providers may vary even for the same firm and year.

## 5.1. Bloomberg

The Bloomberg ESG Score is designed to provide a standardized measure of a company's environmental, social, and governance characteristics based on publicly disclosed information. Rather than assessing firms against predefined sustainability targets, the score focuses on the extent and consistency of ESG disclosure. The score is expressed on a scale from 0 to 10, with higher values indicating more extensive ESG-related disclosure as assessed within Bloomberg's methodological framework. (Bloomberg, 2025)

The Bloomberg ESG framework is structured around the three ESG pillars—Environmental, Social, and Governance—each of which consists of several thematic categories. These categories cover a broad range of topics, including emissions, resource usage, workforce practices, product responsibility, board structure, and shareholder rights. Bloomberg collects ESG data primarily from publicly available sources such as annual reports, sustainability reports, regulatory filings, and company disclosures. The methodology does not rely on company surveys, but instead prioritizes information that can be independently verified. (Bloomberg, 2025)

A central feature of the Bloomberg ESG Score is its industry-specific design. ESG topics are not considered equally relevant across all sectors; therefore, the selection of metrics and their relative importance varies by industry. This sector-based approach aims to enhance comparability by focusing on ESG issues that are most relevant to firms' core business activities. Companies are thus assessed relative to peers within the same industry rather than against a uniform cross-sector benchmark. (Bloomberg, 2025)

## 5.2. LSEG

Building on the disclosure-focused approach outlined for the Bloomberg ESG Score, this subsection describes the ESG scoring methodology applied by LSEG. While both providers aim to summarize firms' environmental, social, and governance characteristics in a standardized form, LSEG follows a conceptually distinct approach that places greater emphasis on assessing ESG performance and exposure to sustainability-related risks, rather than primarily on disclosure intensity. (London Stock Exchange Group (LSEG), 2024)

The LSEG ESG framework is structured around the three ESG pillars—Environmental, Social, and Governance—which are further divided into multiple categories and themes. Similar to Bloomberg, the LSEG ESG framework covers a broad range of sustainability-related topics across the environmental, social, and governance dimensions, including areas such as greenhouse gas emissions, resource use, workforce practices, human rights, product responsibility, management structure, and shareholder rights. ESG data are likewise drawn from publicly available sources such as company reports, regulatory filings, and sustainability disclosures. However, LSEG places greater emphasis on the continuous monitoring and reassessment of this information within its scoring process. (London Stock Exchange Group (LSEG), 2024)

A key feature of the LSEG methodology is its distinction between ESG performance and ESG controversies. ESG scores are adjusted to reflect material controversies related to environmental, social or governance issues, allowing the scoring system to account for adverse events that may not be fully captured by standard disclosures alone. This mechanism aims to provide a more comprehensive view of a firm's ESG profile by incorporating both reported practices and observed incidents. (London Stock Exchange Group (LSEG), 2024)

The LSEG ESG Score is calculated on a 0 to 100 scale, with higher values indicating stronger ESG performance relative to industry peers. Similar to Bloomberg, firms are assessed in relation to comparable companies operating within the same sector, which supports peer-based comparability. (London Stock Exchange Group (LSEG), 2024)

# 6. Variables and Descriptive Statistics

In this chapter, we introduce the economic variables used to analyse the effects of ESG scores on a company's financial performance. To assess potential causal relationships, we employ regression analysis. Regressions consist of dependent and explanatory variables, and the objective is to estimate the relationship between them. In this context, the selection of variables is crucial, as it determines whether the results are economically meaningful. In the following sections, we present the dependent variables as well as the explanatory variables, namely the ESG scores.

## 6.1. Dependent Variables

The aim of this paper is to examine the effects of ESG scores on firms' financial performance. However, financial performance is a broad concept that can be interpreted differently depending on the context. To obtain balanced and robust results, we consider four distinct dimensions of corporate performance, which together provide a comprehensive perspective on the potential effects of ESG scores. These four dimensions are valuation, profitability, return, and volatility, represented by the financial indicators book-to-market ratio, return on common equity, quarterly return, and 260-day return volatility. Each of these variables is discussed in detail below.

### 6.1.1. Book-to-Market Ratio

We use the book-to-market ratio as a proxy for firm valuation. Valuation is an important metric, as it reflects investor expectations and sentiment regarding a company's prospects. It can also serve as an indicator of a firm's overall financial condition and perceived future trajectory. The book-to-market ratio is defined as the book value of equity divided by the market value of equity:

$$\mathrm{B/M}_{i,t} = \frac{\text{Book Value of Equity}_{i,t}}{\text{Market Value of Equity}_{i,t}}. \tag{6.1}$$

As shown in the formula above, the book-to-market ratio is the inverse of the price-to-book ratio. We use the former because it is more commonly applied in empirical asset pricing research and often yields more robust results.(Fama and French, 1995)(Fama and French, 2015)

### 6.1.2. Volatility

Volatility is a well-established measure in empirical finance and has been widely used in asset pricing research (Ang et al., 2006). In this study, we use 260-day return volatility as a proxy for firm-level risk. The underlying hypothesis is that higher ESG scores—particularly in the governance dimension—are associated with safer corporate practices, which may reduce firm-specific risk and, consequently, return volatility.

### 6.1.3. Quarterly Returns

Stock returns are a central measure of firm performance, as they reflect value creation for shareholders. If higher ESG scores contribute to more sustainable or resilient business models, this may be reflected in quarterly returns. Moreover, quarterly returns are frequently used in empirical asset pricing studies and therefore provide comparability with existing literature (Fama and French, 1992).

### 6.1.4. Profitability (ROCE)

Profitability plays a central role in the five-factor asset pricing model proposed by Fama and French (Fama and French, 2015). Empirical evidence suggests that firms with higher operating profitability systematically outperform firms with lower profitability. It is therefore essential to include a profitability measure in our analysis.

Return on common equity (ROCE) is defined as earnings before interest and taxes (EBIT) divided by common equity, where common equity equals total assets minus total liabilities and preferred equity. ROCE measures how efficiently a firm generates profits from shareholders' invested capital and thus provides a meaningful indicator of operational performance.

Table 6.1.: Summary statistics for the dependent variables and ESG scores. Financial variables are pooled across sectors, while ESG scores are reported separately for banking and energy firms due to sector-specific score construction.

| Variable | Sample | Obs | Mean | Median | Std. Dev. | Min | Max |
|---|---|---|---|---|---|---|---|
| Book-to-Market | Full sample | 3108 | 1.5589 | 1.0801 | 2.9721 | 0.0086 | 95.2381 |
| Volatility (260) | Full sample | 3128 | 37.2196 | 34.2685 | 15.5281 | 2.7760 | 188.3810 |
| Return (Q1) | Full sample | 3146 | 0.0190 | 0.0181 | 0.1882 | −0.7505 | 1.8577 |
| ROCE | Full sample | 3114 | 6.3147 | 7.8328 | 18.6262 | −277.7234 | 136.6026 |
| Bloomberg ESG | Banks | 994 | 3.4289 | 3.2300 | 1.2352 | 1.0100 | 7.8500 |
| Bloomberg ESG | Energy | 595 | 5.0599 | 5.2700 | 1.3457 | 1.4100 | 7.3200 |
| LSEG ESG | Banks | 1480 | 67.1690 | 72.2181 | 19.7251 | 3.0850 | 95.6637 |
| LSEG ESG | Energy | 1008 | 75.2884 | 77.9404 | 12.3735 | 26.6965 | 93.1678 |

## 6.2. ESG Variables

The explanatory variables consist of two sets of ESG scores: one provided by Bloomberg and one by LSEG. There is no universally accepted methodology for constructing ESG scores, and substantial differences exist across providers. As illustrated in the figure below, the two ESG datasets differ in several respects, including their scale (0–10 for Bloomberg and 0–100 for LSEG) and their development over time.

In particular, Bloomberg ESG scores exhibit a clear upward trend over time, with both mean and median values increasing throughout the sample period. While this pattern may reflect a genuine improvement in ESG compliance among firms, it may also capture market-wide trends in disclosure practices or rating methodologies. These time trends must therefore be taken into account and appropriately controlled for in the regression analysis.

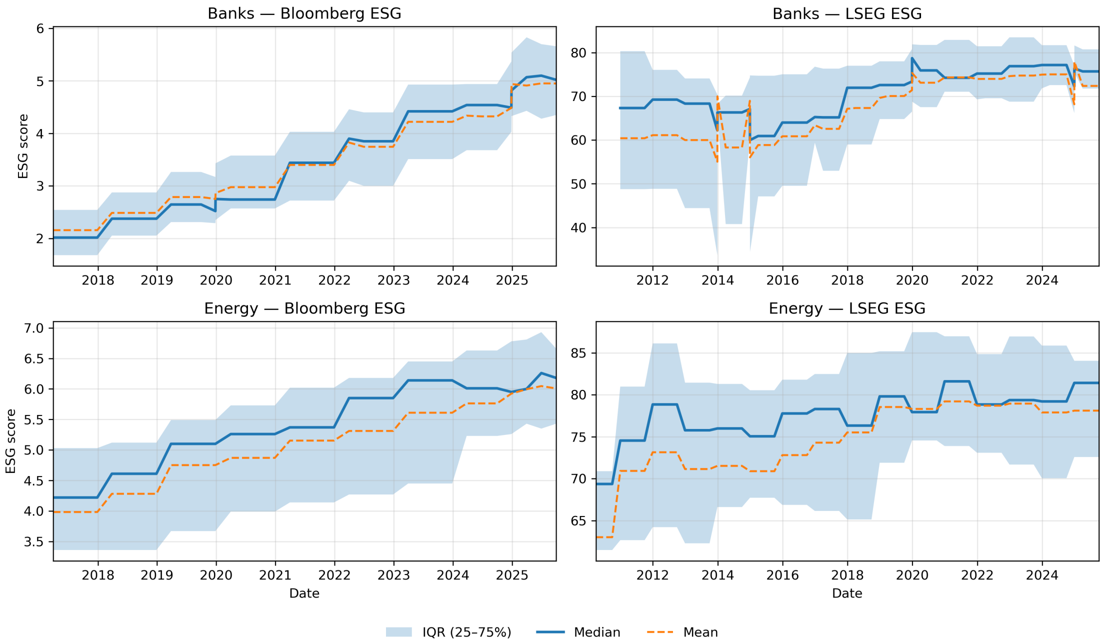


Figure 6.1.: Time evolution and cross-sectional dispersion of Bloomberg and LSEG ESG scores for European banking and energy firms. The figure displays the median, mean, and interquartile range across firms.

# 7. Empirical Methodology

In this chapter, we define the methodology used to conduct our empirical study. As discussed in the previous chapters, our dataset consists of a large number of observations from multiple firms across different sectors (banks and energy companies) and ESG providers (Bloomberg and LSEG) over an extended time period. A regression framework is the most suitable approach to examine the relationship between an explanatory variable and a dependent variable, and it has also been adopted in similar prior studies (Anselmi and Petrella, 2025).

Since our data covers multiple firms over time, resulting in a two-dimensional structure, a panel regression framework is appropriate to obtain robust results. In the following sections, we describe the four regression specifications employed in the panel setting. Given the clear methodological differences across ESG providers and sectors, we maintain this structure and estimate each regression separately for one ESG provider and one sector at a time.

Furthermore, we include control variables. Control variables are factors that are correlated with the dependent variable $Y$. By incorporating these controls, we aim to isolate the effect of ESG scores on $Y$ from influences attributable to other determinants. The choice of control variables depends on the specific dependent variable considered. The detailed reasoning behind each selection is discussed in the subsequent chapter. We now turn to the four fundamental regression specifications used in this study.

## 7.1. Pooled OLS

Pooled OLS is the most basic regression specification employed in this study. In this framework, all observations across firms and time periods are treated as a single dataset, and the relationship between the ESG score and the dependent variable $Y$ is estimated without accounting for firm or time heterogeneity. The specification is given by:

$$Y_{i,t} = \alpha + \beta\,\mathrm{ESG}_{i,t} + \mathbf{X}'_{i,t}\boldsymbol{\gamma} + \varepsilon_{i,t}, \tag{7.1}$$

where $\mathbf{X}_{i,t}$ denotes the vector of control variables.

The pooled OLS regression does not isolate the effect of ESG scores from unobserved firm-specific or time-specific influences and therefore does not allow for causal interpretation. Nevertheless, it provides a useful benchmark and identifies baseline correlations that can later be compared with more restrictive specifications. A key limitation of pooled OLS is that it ignores time-invariant firm-specific characteristics and common market-wide trends, both of which may bias the estimated ESG coefficient. The following sections introduce methods to account for these effects.

## 7.2. Firm Fixed Effects

Firm fixed effects (FE) control for time-invariant, firm-specific characteristics that may influence the dependent variable. Mathematically, we introduce the term $\mu_i$, which captures unobserved characteristics that remain constant for each firm:

$$Y_{i,t} = \alpha + \beta\,\mathrm{ESG}_{i,t} + \mathbf{X}'_{i,t}\gamma + \mu_i + \varepsilon_{i,t}. \tag{7.2}$$

The results of this regression indicate how within-firm changes in ESG scores are associated with changes in the dependent variable. A positive coefficient $\beta$ implies that an increase in a firm's ESG score is associated with an increase in the dependent variable for that firm. However, firm fixed effects do not account for common temporal trends affecting all firms simultaneously.

## 7.3. Time Fixed Effects

To determine whether the relationship between ESG scores and the dependent variable $Y$ is driven by market-wide trends rather than firm-level ESG changes, we introduce time fixed effects. The logic is analogous to firm fixed effects, but the correction operates along the time dimension instead of the firm dimension. The time-specific term $\lambda_t$ absorbs all effects that are constant across firms within a given time period. The specification is therefore:

$$Y_{i,t} = \alpha + \beta\,\mathrm{ESG}_{i,t} + \mathbf{X}'_{i,t}\gamma + \lambda_t + \varepsilon_{i,t}. \tag{7.3}$$

A significant coefficient $\beta$ after controlling for time fixed effects suggests that the relationship between ESG scores and the dependent variable is not solely driven by macroeconomic or market-wide developments. However, this specification does not control for persistent firm-specific characteristics.

## 7.4. Two-Way Fixed Effects

To account simultaneously for firm-specific and time-specific unobserved heterogeneity, we estimate a two-way fixed effects model. This specification combines firm and time fixed effects:

$$Y_{i,t} = \alpha + \beta\,\mathrm{ESG}_{i,t} + \mathbf{X}'_{i,t}\gamma + \mu_i + \lambda_t + \varepsilon_{i,t}. \tag{7.4}$$

This is the most restrictive specification and provides the most informative estimates regarding the effect of ESG scores on the dependent variable. By controlling for both firm-specific and market-wide effects, the estimated coefficient $\beta$ reflects within-firm variation over time, net of common trends. This approach is widely used in the literature as a standard framework for panel data analysis (Albuquerque et al., 2020)(Bolton and Kacperczyk, 2021).

## 7.5. Interpretation

The four regression specifications are not substitutes but complementary tools. While the existence of an effect between ESG scores and the dependent variable is primarily assessed using the two-way fixed effects model, the other specifications provide valuable insights into the structure of the relationship.

For example, a larger coefficient in the firm fixed effects model relative to the two-way fixed effects model may indicate that part of the association is driven by market-wide trends. Conversely, a larger coefficient in pooled OLS compared to firm fixed effects suggests that cross-sectional differences between firms, rather than within-firm ESG changes, account for much of the observed relationship.

Even when the two-way fixed effects model provides the main reference point, the comparison across specifications helps to disentangle structural firm characteristics, time trends, and within-firm dynamics. In many cases, particularly after controlling for both firm and time fixed effects, coefficients may lose statistical significance. In such situations, examining the less restrictive models can help to better understand the underlying patterns in the data.

In conclusion, we estimate four regression specifications across ESG providers and sectors to identify and interpret the relationship between ESG scores and firms' financial performance.

# 8. Results and Discussion

In this chapter, we present and discuss our empirical findings and examine their implications in the broader context of ESG scores and firms' overall financial performance. To assess financial performance, we consider multiple dimensions, including firm valuation, profitability, volatility, and returns. To obtain a more balanced view, we conduct the analysis for two sectors (banking and energy) and use two ESG score providers (Bloomberg and LSEG). Finally, we summarize and compare the results across providers and across sectors.

For the descriptive statistics in Chapter 6, we reported ESG scores in provider units to preserve interpretability. In this chapter, we use standardized ESG scores (z-scores), since the focus is on cross-provider and cross-sector comparison. Coefficients and visual slopes therefore represent the effect of a one-standard-deviation change in ESG. All results are provided in both standardized units and original ESG units in Appendix A.

## 8.1. Valuation Effects

We use the book-to-market ratio to measure firm valuation. A high book-to-market ratio indicates that the market values the company relatively less, whereas a low book-to-market ratio indicates the opposite. In all valuation regressions, we include 260-day volatility and return on common equity as control variables. Lower volatility is preferred by risk-averse investors and has been linked to higher risk-adjusted returns and, consequently, higher valuation in empirical studies (Campbell and Shiller, 1988). Firms with higher ROCE (return on common equity) have also been found to trade at higher price-to-book ratios and, as a result, lower book-to-market ratios (Fama and French, 2015). By controlling for these relationships, we better isolate the association between ESG scores and valuation.

For LSEG, we find a small positive effect on the book-to-market ratio. In banks, pooled OLS indicates a general correlation between ESG scores and valuation, whereas this general pattern is not observed in the energy sector. However, after controlling for firm fixed effects, we find a significant positive coefficient, indicating that an increase in a firm's ESG score is associated with a higher book-to-market ratio and therefore a lower valuation. Notably, the estimated coefficient is roughly an order of magnitude larger for banks than for energy firms.

Sectoral differences are also evident in the two-way fixed effects (TWFE) specifications, although the estimated effects are smaller in magnitude. Importantly, the change from firm FE to TWFE goes in opposite directions across sectors. For banks, the coefficient decreases under TWFE, whereas it increases slightly for energy firms. For banks, this suggests that part of the effect under firm FE reflects a positive time trend in ESG scores that is correlated with book-to-market ratios. Once both time and firm fixed effects are included, the estimated effect is therefore diminished. Although the time fixed effects results are not statistically significant, the positive sign of the time FE coefficient is consistent with this interpretation.

For energy firms, the pattern is reversed: the coefficient is higher when controlling for both time and firm fixed effects. This suggests that time trends are negatively correlated with ESG scores and book-to-market ratios. In other words, over time, increasing ESG scores coincide with lower book-to-market ratios (higher valuation) at the market level; however, within a given firm, increases in ESG are associated with higher book-to-market ratios (lower valuation), as reflected by the positive coefficients in both the firm FE and TWFE specifications.

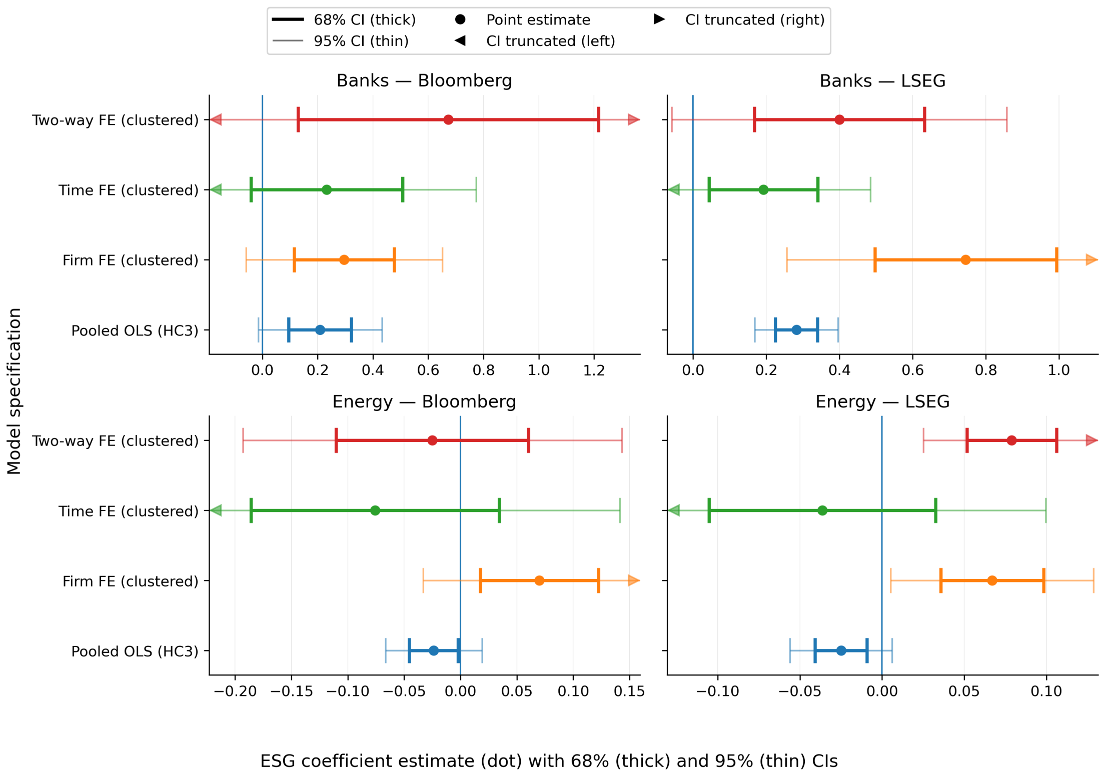


Figure 8.1.: Coefficient estimates of ESG scores on firm valuation (book-to-market ratio) across alternative panel regression specifications. Dots indicate point estimates; whiskers denote 68% (thick) and 95% (thin) confidence intervals. Results are shown separately for banks and energy firms, and for Bloomberg and LSEG ESG providers. All regressions include return on common equity and volatility as control variables.

Although the pure time fixed effects results are again not statistically significant, the negative sign of the coefficient is consistent with this interpretation.

For Bloomberg, the regressions yield mostly non-significant results, with the exception of pooled OLS in banks. The positive coefficient suggests that higher ESG scores are associated with a higher book-to-market ratio, i.e., lower valuation. However, since neither firm nor time fixed effects produce significant coefficients, the relationship between Bloomberg ESG scores and the book-to-market ratio appears to be driven primarily by cross-sectional differences and/or common time trends rather than within-firm ESG changes.

Overall, the results suggest a negative association between ESG scores and firm valuation (i.e., higher book-to-market ratios). In comparison, LSEG scores capture within-firm effects on valuation more clearly, whereas Bloomberg primarily reflects broader correlations, if any. In addition, the magnitude of the estimated effects appears larger for banks than for energy firms.

## 8.2. Return Effects

To estimate the effects of ESG scores on returns, we use quarterly returns as the dependent variable and include two control variables: book-to-market ratio and 260-day volatility. These controls account for the established relationships between returns, valuation, and volatility. Regarding volatility, Ang et al. document a negative relation between volatility and average returns (Ang et al., 2006). By controlling for these dependencies, we aim to better isolate the association between ESG scores and returns.

For LSEG, we do not find evidence of a relationship between ESG scores and returns: all specifications yield non-significant results in both sectors, and this conclusion is unchanged when controlling for firm or time fixed effects. For Bloomberg, the picture differs. In banks, we find a positive coefficient in pooled OLS and an even larger coefficient (around 3%) when controlling for firm fixed effects. However, once time fixed effects are included, the coefficients become non-significant. This suggests that Bloomberg ESG scores are primarily correlated with market-wide trends that are absorbed by time fixed effects.

In the energy sector, the relationship is reversed. Pooled OLS and firm fixed effects yield non-significant results, implying neither a cross-sectional ESG return premium nor a simple within-firm association. However, under TWFE we obtain a negative coefficient, suggesting that ESG improvements beyond the market-wide trend are associated with lower returns. Comparing the two sectors, this pattern may reflect sector-specific dynamics. For banks, periods of higher returns coincide with market-wide improvements in ESG, while within-firm ESG improvements beyond common trends are not rewarded. For energy firms, there is no clear market-wide trend in ESG, but firm-specific ESG improvements relative to peers are associated with lower short-run returns.

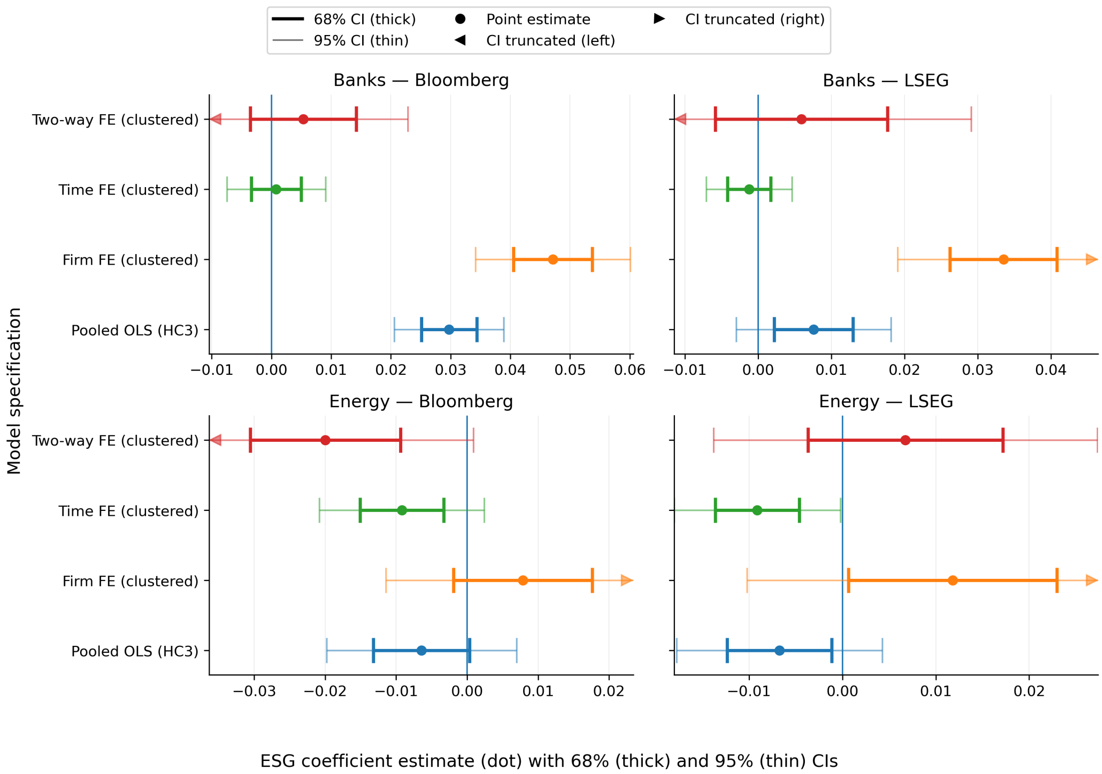


Figure 8.2.: Estimated ESG coefficients on quarterly stock returns.

## 8.3. Profitability Effects

We use ROCE as a measure of firm profitability. In these regressions, we include two control variables: book-to-market ratio as a valuation proxy and 260-day volatility as a risk proxy. The relationship between valuation and profitability has already been discussed in the previous chapter. In addition, the five-factor asset pricing model by Fama and French links higher operating profitability to more robust return characteristics (Fama and French, 2015). Controlling for volatility and valuation therefore helps to remove variation that is not attributable to ESG scores and yields a more informative assessment of the ESG–profitability relationship.

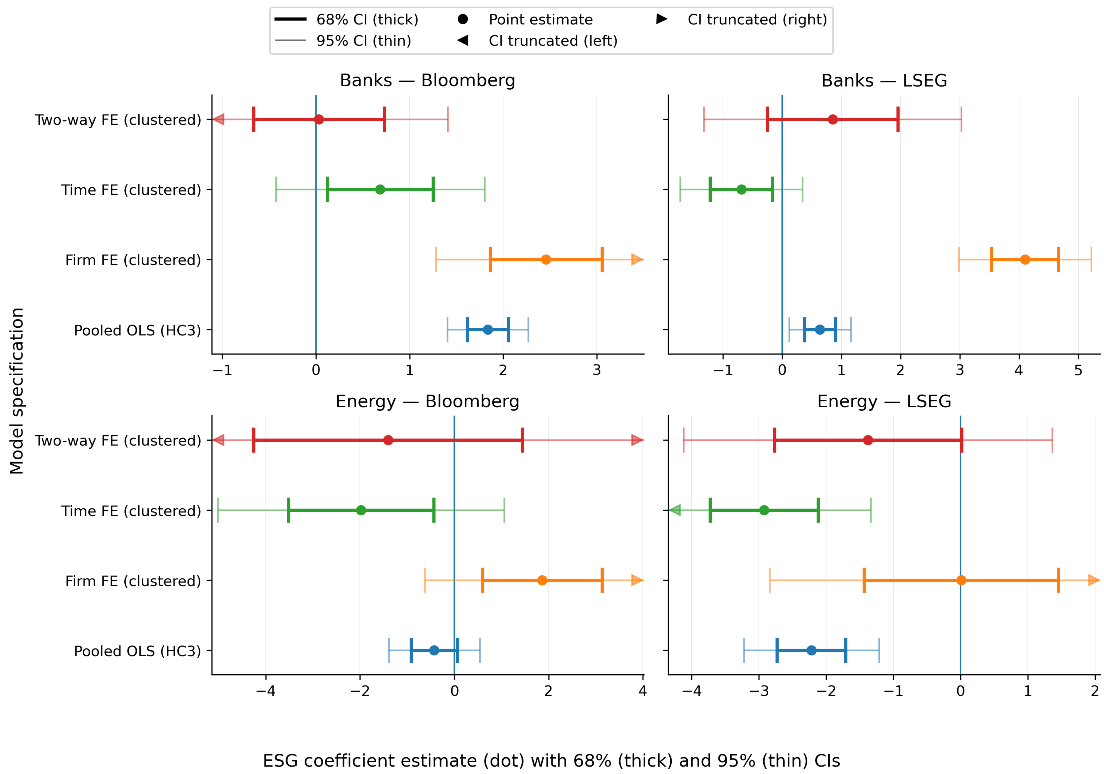


Figure 8.3.: Estimated ESG coefficients on firm profitability (return on common equity).

Overall, there are substantial differences across sectors. For banks, we obtain similar patterns for both ESG providers. Bloomberg and LSEG both yield significant positive coefficients in pooled OLS and firm fixed effects regressions, whereas specifications with time fixed effects are non-significant. Although the Bloomberg coefficients are at least an order of magnitude larger, the direction is consistent across providers. Positive pooled OLS coefficients indicate that banks with higher ESG scores tend to exhibit higher profitability. The larger coefficients under firm fixed effects further suggest that within-bank ESG improvements are associated with higher ROCE. However, since these effects disappear once time fixed effects are included, the results indicate that much of the variation reflects macro trends. This suggests that ESG and ROCE move together primarily due to market-wide forces (e.g., regulatory changes, macroeconomic conditions, or broad sector developments) rather than firm-specific ESG initiatives.

In contrast, the energy sector yields a different pattern. For Bloomberg, we do not obtain significant results, suggesting no detectable relationship between Bloomberg ESG scores and profitability in energy firms. For LSEG, we find a negative coefficient in pooled OLS and in the time fixed effects specification, but the results become non-significant once firm fixed effects are included. The pooled OLS result suggests that energy firms with higher ESG scores tend to have lower profitability. Since the relationship disappears under firm fixed effects, changes in ESG within a given firm do not appear to be significantly associated with changes in profitability. The time fixed effects results indicate that the association is largely driven by firm-specific structural differences rather than by within-firm ESG changes. Overall, the ESG–profitability relationship exhibits strong sectoral differences: in banking, ESG scores move positively with profitability primarily through market-wide trends, whereas in energy, LSEG scores appear to separate firms based on structural differences that are correlated with profitability.

## 8.4. Volatility Effects

To estimate the effects of ESG scores on volatility, we use 260-day return volatility as the dependent variable and include two control variables: book-to-market ratio (valuation) and return on common equity (profitability). The relationships between volatility, book-to-market ratio, and ROCE have been established in the previous chapters. Controlling for these variables allows for a more accurate assessment of the association between ESG scores and return volatility.

For Bloomberg, we do not obtain significant results in either sector. For LSEG, pooled OLS yields a small negative coefficient for both banks and energy firms, indicating that higher ESG scores are associated with lower volatility. While adding firm or time fixed effects yields non-significant results for banks, the time fixed effects specification yields a stronger negative coefficient for energy firms. Overall, this suggests that LSEG ESG scores are associated with lower return volatility. However, the sources of this association differ across sectors. For banks, the correlation appears to be driven by cross-sectional differences and market-wide trends. For energy firms, the time fixed effects results suggest that structural differences across firms play a larger role. In other words, energy firms with higher ESG scores may be fundamentally different from their peers in ways that lead to lower volatility. Consistent with the firm fixed effects results, improving ESG within a given firm does not necessarily reduce volatility, as no within-firm relationship is identified once firm fixed effects are included.

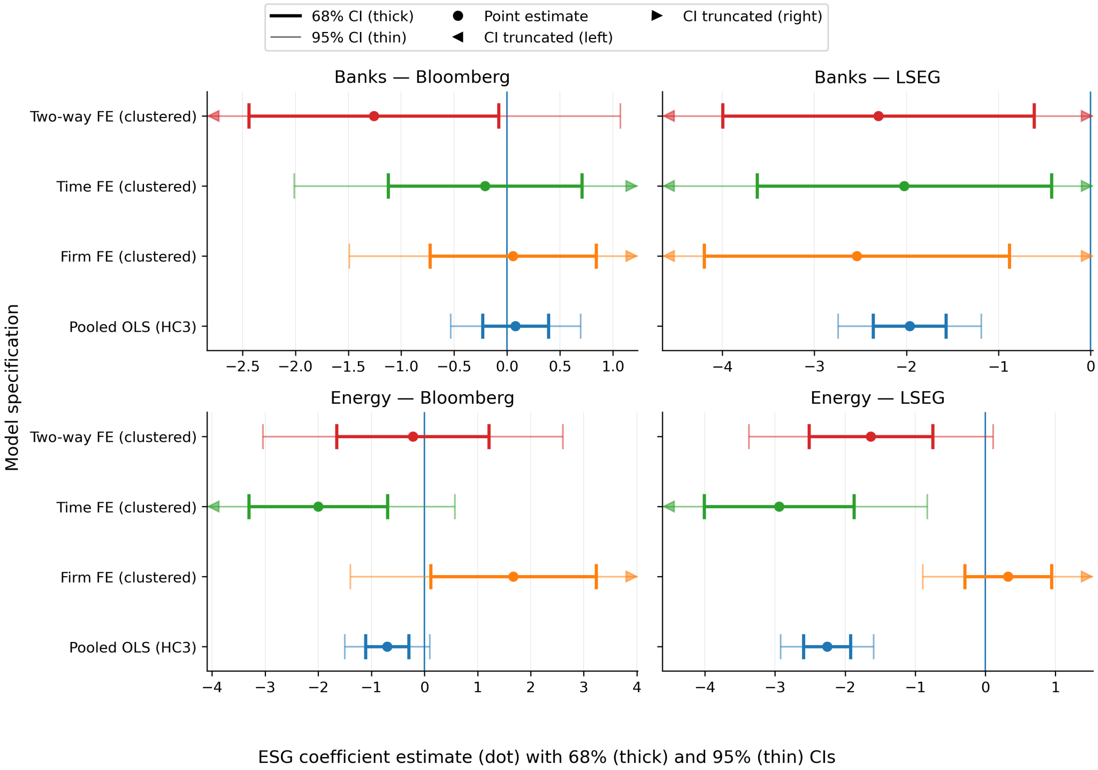


Figure 8.4.: Estimated ESG coefficients on stock return volatility.

## 8.5. Comparison ESG Providers

As discussed in the previous chapters, ESG metrics are not uniform and can differ substantially across providers. These differences partly reflect deliberate design choices, as providers target different users and differentiate their products through distinct methodologies. In our results, these methodological differences translate into differences in the relationships between ESG scores and firms' financial performance.

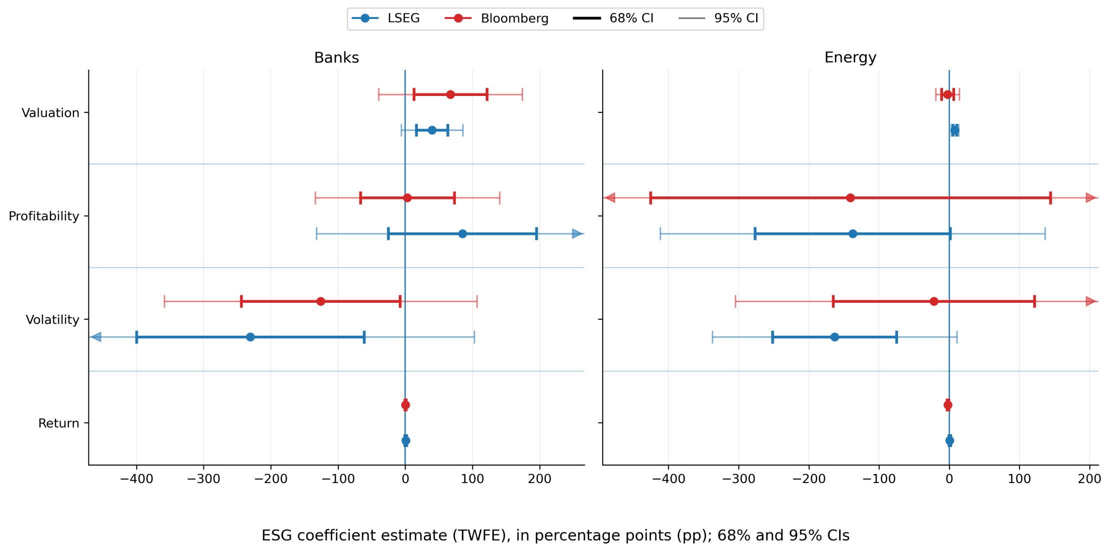


Figure 8.5.: Provider comparison of ESG coefficients across outcomes under two-way fixed effects.

Overall, we do not identify a clear "winner" between Bloomberg and LSEG in terms of capturing relationships with financial performance. LSEG yields significant results in more categories, but Bloomberg performs better in some dimensions, particularly returns. At the same time, LSEG does not produce significant results for returns in either sector. For valuation, we find a negative association (higher book-to-market ratios), although the results are largely explained by time trends. For profitability, LSEG yields positive associations with ROCE in banks and negative associations in energy firms; however, the TWFE specifications are not statistically significant. In banks, the associations appear mainly driven by macro trends, whereas in energy firms they appear largely driven by structural differences rather than within-firm ESG changes. For volatility, pooled OLS indicates that higher LSEG ESG scores are associated with lower volatility in both sectors, but the results again reflect structural differences (energy) and a combination of market-wide and cross-sectional effects (banks).

In contrast, Bloomberg primarily identifies relationships in pooled OLS and mainly for banks. It indicates correlations between higher ESG scores and lower valuation, higher returns, and higher profitability in banks, but these patterns are largely explained by cross-sectional differences and market-wide trends (and, for returns, primarily by market-wide trends). In addition, Bloomberg suggests a relationship between higher ESG scores and lower returns in energy firms under TWFE. Beyond this, we do not obtain significant results for energy firms in the other categories, and Bloomberg ESG scores do not exhibit a robust relationship with volatility.

Overall, our results highlight substantial differences between Bloomberg and LSEG ESG scores. Both providers perform differently across outcome variables. In general, LSEG is more versatile and yields significant results in more settings than Bloomberg.

## 8.6. Comparison Across Sectors

As established in the previous chapters, ESG scores are sector-dependent: providers use different metrics and weights across industries. This naturally affects the relationship between ESG scores and financial performance outcomes, alongside other sector-specific characteristics such as differing market dynamics and regulatory pressures.

Regarding valuation, we identify a clear negative association between ESG scores and valuation among banks, whereas no comparable effect is observed for the energy sector. In other cases, we obtain significant results in both sectors but with opposite signs. This occurs for returns, where we find an overall positive market-wide association between Bloomberg ESG scores and returns in banking, while TWFE yields a negative effect of Bloomberg ESG scores on returns in energy. In yet other cases, coefficients have the same sign across sectors but appear to reflect different mechanisms. For volatility, LSEG ESG scores exhibit an overall negative association with volatility, but the evidence suggests that the relationship is mainly driven by cross-sectional differences and market-wide trends for banks, whereas in energy firms it is primarily driven by structural differences. For profitability, we find positive associations in banks and negative associations in energy.

These results indicate that ESG scores vary not only across providers but also across sectors. Some differences can be attributed to sector-specific score construction, while others likely reflect fundamental differences in how sectors operate.

In conclusion, we do not find a strong and consistent linkage between ESG scores and firms' financial performance. While ESG scores may correlate with certain financial metrics, these correlations often weaken or disappear once firm and/or time fixed effects are included. This suggests that ESG scores do not systematically explain financial performance in a way that is solely attributable to ESG itself. Instead, ESG scores tend to co-move with macro developments (e.g., expansions and recessions, political sentiment regarding ESG) or capture firm classifications that coincide with structural characteristics that are the actual drivers of performance. Overall, ESG scores have limited explanatory power in this setting, and improving ESG scores to improve short-run financial performance is likely to have, at best, a modest effect. Furthermore, the lack of standardization across ESG providers makes it difficult to identify consistent patterns across providers or sectors. Accordingly, ESG scores should be evaluated separately by provider and sector, and conclusions should not be directly transferred across contexts.

# 9. Conclusion

This thesis examined whether ESG scores are systematically related to firm-level financial performance and whether these relationships differ across sectors and ESG data providers.

Overall, the results do not suggest a causal relationship between ESG scores and firms' financial performance. However, we identify significant correlations in several areas. These correlations are largely attributable to cross-sectional differences (particularly for profitability and volatility) and to market-wide trends (particularly for returns). In addition, we find substantial differences across both ESG providers and sectors. These differences often manifest either as correlations appearing for one provider but not the other, or as correlations of opposite sign across sectors. In general, ESG scores exhibit a modest negative correlation with valuation and volatility, while profitability is positively correlated with ESG scores in banks but negatively correlated in energy companies.

This thesis provides a baseline for comparing the relationship between ESG scores and firm financial performance by offering a detailed comparison across two ESG providers and two sectors. The analysis highlights several structural issues regarding ESG scores; nevertheless, a more comprehensive study could expand the number of providers and sectors considered. Furthermore, this paper focuses on short-term relationships between ESG scores and financial performance. While we do not find evidence of a short-run causal relationship, it would be useful to examine long-run effects, particularly given that ESG metrics are intended to capture aspects related to sustainable growth.

In addition, this thesis does not assess the individual effects of the three ESG pillars. While the aggregated ESG score does not display strong causal relationships with financial performance, the individual pillars may exhibit different patterns. Future research could therefore analyse pillar-level effects and, more granularly, investigate the role of specific components within each pillar.

Overall, this thesis establishes a robust baseline for assessing the relationship between ESG scores and financial performance. It also highlights the heterogeneity of ESG scores across providers and sectors. Finally, the coding framework was developed with extensibility in mind, allowing future studies to build upon it.

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# Declaration of AI–Assisted Technologies in the Writing Process

During the preparation of this thesis, language-support tools such as ChatGPT were used to improve clarity and readability. All content was subsequently reviewed and revised by the author, who takes full responsibility for the accuracy and integrity of the work.

# A. Appendix 1 – Regression Results

This appendix presents the detailed results of all regressions conducted in this thesis. The regression tables are organized by dependent variable (book-to-market ratio, ROCE, quarterly return, and volatility). Each table reports four model specifications (pooled OLS, firm fixed effects, time fixed effects, and two-way fixed effects) for both ESG providers and both sectors. Results are presented for raw ESG scores as well as standardized ESG scores (z-scores).

**General Notes on Regression Tables.** All regression tables report coefficient estimates, 68% and 95% confidence intervals, and p-values. Statistical significance at the 5% level is indicated by **. Unless otherwise stated, regressions include controls for the book-to-market ratio and 260-day volatility.

## A. Appendix 1 – Regression Results

Table A.1.: Regression Results for Book-to-Market Ratio (Raw ESG; 68% and 95% CI; Controls in Volatility and ROCE)

| Provider | Model | $\beta$ | Sig | CI68 Low | CI68 High | CI95 Low | CI95 High | p-value | N |
|---|---|---|---|---|---|---|---|---|---|
| **Banks** | | | | | | | | | |
| Bloomberg | Pooled OLS | 0.1691 | | 0.0771 | 0.2610 | −0.0122 | 0.3503 | 0.0675 | 989 |
| Bloomberg | Firm FE | 0.2398 | | 0.0937 | 0.3858 | −0.0481 | 0.5276 | 0.1029 | 989 |
| Bloomberg | Time FE | 0.1885 | | −0.0336 | 0.4105 | −0.2491 | 0.6260 | 0.3988 | 989 |
| Bloomberg | Two-way FE | 0.5443 | | 0.1045 | 0.9840 | −0.3224 | 1.4110 | 0.2187 | 989 |
| LSEG | Pooled OLS | 0.0143 | ** | 0.0114 | 0.0172 | 0.0085 | 0.0201 | 0.0000 | 1474 |
| LSEG | Firm FE | 0.0378 | ** | 0.0252 | 0.0503 | 0.0130 | 0.0626 | 0.0029 | 1474 |
| LSEG | Time FE | 0.0097 | | 0.0022 | 0.0173 | −0.0051 | 0.0246 | 0.1989 | 1474 |
| LSEG | Two-way FE | 0.0203 | | 0.0085 | 0.0320 | −0.0029 | 0.0435 | 0.0873 | 1474 |
| **Energy** | | | | | | | | | |
| Bloomberg | Pooled OLS | −0.0177 | | −0.0338 | −0.0015 | −0.0495 | 0.0142 | 0.2771 | 595 |
| Bloomberg | Firm FE | 0.0520 | | 0.0131 | 0.0909 | −0.0246 | 0.1287 | 0.1840 | 595 |
| Bloomberg | Time FE | −0.0563 | | −0.1382 | 0.0255 | −0.2177 | 0.1050 | 0.4941 | 595 |
| Bloomberg | Two-way FE | −0.0185 | | −0.0820 | 0.0449 | −0.1435 | 0.1064 | 0.7714 | 595 |
| LSEG | Pooled OLS | −0.0020 | | −0.0033 | −0.0007 | −0.0045 | 0.0005 | 0.1158 | 997 |
| LSEG | Firm FE | 0.0054 | ** | 0.0029 | 0.0080 | 0.0004 | 0.0104 | 0.0334 | 997 |
| LSEG | Time FE | −0.0029 | | −0.0085 | 0.0026 | −0.0139 | 0.0081 | 0.6007 | 997 |
| LSEG | Two-way FE | 0.0064 | ** | 0.0042 | 0.0086 | 0.0020 | 0.0107 | 0.0040 | 997 |

Table A.2.: Regression Results for Book-to-Market Ratio (Standardized ESG Z-scores; 68% and 95% CI; Controls in Volatility and ROCE)

| Provider | Model | $\beta$ | Sig | CI68 Low | CI68 High | CI95 Low | CI95 High | p-value | N |
|---|---|---|---|---|---|---|---|---|---|
| **Banks** | | | | | | | | | |
| Bloomberg | Pooled OLS | 0.2089 | | 0.0953 | 0.3225 | −0.0150 | 0.4327 | 0.0675 | 989 |
| Bloomberg | Firm FE | 0.2962 | | 0.1158 | 0.4766 | −0.0594 | 0.6517 | 0.1029 | 989 |
| Bloomberg | Time FE | 0.2328 | | −0.0415 | 0.5070 | −0.3077 | 0.7733 | 0.3988 | 989 |
| Bloomberg | Two-way FE | 0.6723 | | 0.1291 | 1.2155 | −0.3983 | 1.7429 | 0.2187 | 989 |
| LSEG | Pooled OLS | 0.2824 | ** | 0.2246 | 0.3401 | 0.1686 | 0.3961 | 0.0000 | 1474 |
| LSEG | Firm FE | 0.7449 | ** | 0.4967 | 0.9930 | 0.2558 | 1.2339 | 0.0029 | 1474 |
| LSEG | Time FE | 0.1921 | | 0.0435 | 0.3407 | −0.1008 | 0.4850 | 0.1989 | 1474 |
| LSEG | Two-way FE | 0.3997 | | 0.1674 | 0.6320 | −0.0581 | 0.8576 | 0.0873 | 1474 |
| **Energy** | | | | | | | | | |
| Bloomberg | Pooled OLS | −0.0238 | | −0.0455 | −0.0020 | −0.0666 | 0.0191 | 0.2771 | 595 |
| Bloomberg | Firm FE | 0.0700 | | 0.0177 | 0.1224 | −0.0332 | 0.1732 | 0.1840 | 595 |
| Bloomberg | Time FE | −0.0758 | | −0.1860 | 0.0344 | −0.2929 | 0.1413 | 0.4941 | 595 |
| Bloomberg | Two-way FE | −0.0249 | | −0.1103 | 0.0604 | −0.1931 | 0.1432 | 0.7714 | 595 |
| LSEG | Pooled OLS | −0.0250 | | −0.0407 | −0.0092 | −0.0561 | 0.0061 | 0.1158 | 997 |
| LSEG | Firm FE | 0.0671 | ** | 0.0358 | 0.0984 | 0.0054 | 0.1288 | 0.0334 | 997 |
| LSEG | Time FE | −0.0363 | | −0.1053 | 0.0327 | −0.1723 | 0.0996 | 0.6007 | 997 |
| LSEG | Two-way FE | 0.0789 | ** | 0.0517 | 0.1061 | 0.0252 | 0.1326 | 0.0040 | 997 |

Table A.3.: Regression Results for Return Q1 (Raw ESG; 68% and 95% CI; Controls in btm ratio and volatility)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 0.0241 | ** | 0.0203 | 0.0278 | 0.0167 | 0.0315 | 0.0000 | 989 |
| Bloomberg | Firm FE | 0.0381 | ** | 0.0328 | 0.0435 | 0.0276 | 0.0486 | 0.0000 | 989 |
| Bloomberg | Time FE | 0.0006 | | −0.0027 | 0.0040 | −0.0060 | 0.0073 | 0.8504 | 989 |
| Bloomberg | Two-way FE | 0.0043 | | −0.0029 | 0.0115 | −0.0099 | 0.0185 | 0.5509 | 989 |
| LSEG | Pooled OLS | 0.0004 | | 0.0001 | 0.0007 | −0.0002 | 0.0009 | 0.1594 | 1474 |
| LSEG | Firm FE | 0.0017 | ** | 0.0013 | 0.0021 | 0.0010 | 0.0024 | 0.0000 | 1474 |
| LSEG | Time FE | −0.0001 | | −0.0002 | 0.0001 | −0.0004 | 0.0002 | 0.6820 | 1474 |
| LSEG | Two-way FE | 0.0003 | | −0.0003 | 0.0009 | −0.0009 | 0.0015 | 0.6180 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.0048 | | −0.0098 | 0.0003 | −0.0147 | 0.0052 | 0.3471 | 595 |
| Bloomberg | Firm FE | 0.0058 | | −0.0014 | 0.0131 | −0.0085 | 0.0202 | 0.4254 | 595 |
| Bloomberg | Time FE | −0.0068 | | −0.0112 | −0.0024 | −0.0155 | 0.0018 | 0.1216 | 595 |
| Bloomberg | Two-way FE | −0.0148 | | −0.0227 | −0.0070 | −0.0303 | 0.0006 | 0.0609 | 595 |
| LSEG | Pooled OLS | −0.0005 | | −0.0010 | −0.0001 | −0.0014 | 0.0003 | 0.2301 | 997 |
| LSEG | Firm FE | 0.0010 | | 0.0001 | 0.0019 | −0.0008 | 0.0027 | 0.2933 | 997 |
| LSEG | Time FE | −0.0007 | ** | −0.0011 | −0.0004 | −0.0015 | 0.0000 | 0.0449 | 997 |
| LSEG | Two-way FE | 0.0005 | | −0.0003 | 0.0014 | −0.0011 | 0.0022 | 0.5202 | 997 |

Table A.4.: Regression Results for Return Q1 (Standardized ESG Z-scores; 68% and 95% CI; Controls in btm ratio and volatility)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 0.0297 | ** | 0.0251 | 0.0344 | 0.0206 | 0.0389 | 0.0000 | 989 |
| Bloomberg | Firm FE | 0.0471 | ** | 0.0405 | 0.0537 | 0.0341 | 0.0601 | 0.0000 | 989 |
| Bloomberg | Time FE | 0.0008 | | −0.0034 | 0.0050 | −0.0075 | 0.0090 | 0.8504 | 989 |
| Bloomberg | Two-way FE | 0.0053 | | −0.0036 | 0.0142 | −0.0122 | 0.0228 | 0.5509 | 989 |
| LSEG | Pooled OLS | 0.0076 | | 0.0022 | 0.0129 | −0.0030 | 0.0182 | 0.1594 | 1474 |
| LSEG | Firm FE | 0.0335 | ** | 0.0262 | 0.0409 | 0.0191 | 0.0480 | 0.0000 | 1474 |
| LSEG | Time FE | −0.0012 | | −0.0042 | 0.0017 | −0.0071 | 0.0046 | 0.6820 | 1474 |
| LSEG | Two-way FE | 0.0059 | | −0.0059 | 0.0177 | −0.0173 | 0.0291 | 0.6180 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.0064 | | −0.0132 | 0.0004 | −0.0198 | 0.0070 | 0.3471 | 595 |
| Bloomberg | Firm FE | 0.0078 | | −0.0019 | 0.0176 | −0.0114 | 0.0271 | 0.4254 | 595 |
| Bloomberg | Time FE | −0.0092 | | −0.0151 | −0.0033 | −0.0208 | 0.0024 | 0.1216 | 595 |
| Bloomberg | Two-way FE | −0.0200 | | −0.0306 | −0.0094 | −0.0408 | 0.0009 | 0.0609 | 595 |
| LSEG | Pooled OLS | −0.0067 | | −0.0123 | −0.0012 | −0.0178 | 0.0043 | 0.2301 | 997 |
| LSEG | Firm FE | 0.0118 | | 0.0006 | 0.0230 | −0.0102 | 0.0338 | 0.2933 | 997 |
| LSEG | Time FE | −0.0091 | ** | −0.0136 | −0.0046 | −0.0180 | −0.0002 | 0.0449 | 997 |
| LSEG | Two-way FE | 0.0067 | | −0.0037 | 0.0172 | −0.0138 | 0.0273 | 0.5202 | 997 |

Table A.5.: Regression Results for ROCE (Raw ESG; 68% and 95% CI; Controls in btm ratio and volatility)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 1.4850 | ** | 1.3077 | 1.6623 | 1.1355 | 1.8345 | 0.0000 | 989 |
| Bloomberg | Firm FE | 1.9893 | ** | 1.5055 | 2.4731 | 1.0357 | 2.9428 | 0.0000 | 989 |
| Bloomberg | Time FE | 0.5563 | | 0.0988 | 1.0138 | −0.3455 | 1.4581 | 0.2269 | 989 |
| Bloomberg | Two-way FE | 0.0262 | | −0.5380 | 0.5905 | −1.0859 | 1.1384 | 0.9631 | 989 |
| LSEG | Pooled OLS | 0.0324 | ** | 0.0191 | 0.0458 | 0.0061 | 0.0587 | 0.0157 | 1474 |
| LSEG | Firm FE | 0.2079 | ** | 0.1791 | 0.2367 | 0.1512 | 0.2646 | 0.0000 | 1474 |
| LSEG | Time FE | −0.0349 | | −0.0615 | −0.0083 | −0.0873 | 0.0175 | 0.1916 | 1474 |
| LSEG | Two-way FE | 0.0432 | | −0.0127 | 0.0991 | −0.0669 | 0.1533 | 0.4422 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.3166 | | −0.6813 | 0.0481 | −1.0354 | 0.4021 | 0.3879 | 595 |
| Bloomberg | Firm FE | 1.3860 | | 0.4450 | 2.3270 | −0.4686 | 3.2406 | 0.1435 | 595 |
| Bloomberg | Time FE | −1.4716 | | −2.6169 | −0.3263 | −3.7289 | 0.7857 | 0.2019 | 595 |
| Bloomberg | Two-way FE | −1.0458 | | −3.1629 | 1.0712 | −5.2183 | 3.1266 | 0.6234 | 595 |
| LSEG | Pooled OLS | −0.1793 | ** | −0.2206 | −0.1381 | −0.2606 | −0.0980 | 0.0000 | 997 |
| LSEG | Firm FE | 0.0010 | | −0.1159 | 0.1178 | −0.2293 | 0.2313 | 0.9933 | 997 |
| LSEG | Time FE | −0.2360 | ** | −0.3009 | −0.1711 | −0.3639 | −0.1082 | 0.0003 | 997 |
| LSEG | Two-way FE | −0.1113 | | −0.2237 | 0.0012 | −0.3329 | 0.1104 | 0.3254 | 997 |

Table A.6.: Regression Results for ROCE (Standardized ESG Z-scores; 68% and 95% CI; Controls in btm ratio and volatility)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 1.8343 | ** | 1.6153 | 2.0534 | 1.4027 | 2.2660 | 0.0000 | 989 |
| Bloomberg | Firm FE | 2.4572 | ** | 1.8596 | 3.0548 | 1.2794 | 3.6351 | 0.0000 | 989 |
| Bloomberg | Time FE | 0.6872 | | 0.1220 | 1.2523 | −0.4267 | 1.8011 | 0.2269 | 989 |
| Bloomberg | Two-way FE | 0.0324 | | −0.6646 | 0.7294 | −1.3413 | 1.4062 | 0.9631 | 989 |
| LSEG | Pooled OLS | 0.6395 | ** | 0.3761 | 0.9028 | 0.1205 | 1.1585 | 0.0157 | 1474 |
| LSEG | Firm FE | 4.1006 | ** | 3.5330 | 4.6682 | 2.9819 | 5.2193 | 0.0000 | 1474 |
| LSEG | Time FE | −0.6891 | | −1.2137 | −0.1646 | −1.7230 | 0.3447 | 0.1916 | 1474 |
| LSEG | Two-way FE | 0.8519 | | −0.2503 | 1.9541 | −1.3204 | 3.0242 | 0.4422 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.4261 | | −0.9168 | 0.0647 | −1.3933 | 0.5411 | 0.3879 | 595 |
| Bloomberg | Firm FE | 1.8651 | | 0.5989 | 3.1314 | −0.6305 | 4.3608 | 0.1435 | 595 |
| Bloomberg | Time FE | −1.9803 | | −3.5216 | −0.4390 | −5.0180 | 1.0574 | 0.2019 | 595 |
| Bloomberg | Two-way FE | −1.4073 | | −4.2562 | 1.4415 | −7.0221 | 4.2074 | 0.6234 | 595 |
| LSEG | Pooled OLS | −2.2186 | ** | −2.7291 | −1.7082 | −3.2246 | −1.2127 | 0.0000 | 997 |
| LSEG | Firm FE | 0.0121 | | −1.4339 | 1.4582 | −2.8378 | 2.8621 | 0.9933 | 997 |
| LSEG | Time FE | −2.9204 | ** | −3.7231 | −2.1176 | −4.5025 | −1.3382 | 0.0003 | 997 |
| LSEG | Two-way FE | −1.3768 | | −2.7682 | 0.0146 | −4.1191 | 1.3655 | 0.3254 | 997 |

Table A.7.: Regression Results for Volatility (260-day; Raw ESG; 68% and 95% CI; Controls in ROCE and btm ratio)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 0.0634 | | −0.1886 | 0.3153 | −0.4332 | 0.5599 | 0.8025 | 989 |
| Bloomberg | Firm FE | 0.0447 | | −0.5906 | 0.6801 | −1.2075 | 1.2969 | 0.9442 | 989 |
| Bloomberg | Time FE | −0.1699 | | −0.9105 | 0.5706 | −1.6294 | 1.2896 | 0.8196 | 989 |
| Bloomberg | Two-way FE | −1.0197 | | −1.9757 | −0.0637 | −2.9038 | 0.8644 | 0.2891 | 989 |
| LSEG | Pooled OLS | −0.0997 | ** | −0.1197 | −0.0797 | −0.1392 | −0.0603 | 0.0000 | 1474 |
| LSEG | Firm FE | −0.1288 | | −0.2128 | −0.0447 | −0.2945 | 0.0369 | 0.1279 | 1474 |
| LSEG | Time FE | −0.1026 | | −0.1836 | −0.0216 | −0.2622 | 0.0570 | 0.2077 | 1474 |
| LSEG | Two-way FE | −0.1169 | | −0.2026 | −0.0311 | −0.2859 | 0.0522 | 0.1757 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.5215 | | −0.8236 | −0.2194 | −1.1168 | 0.0738 | 0.0860 | 595 |
| Bloomberg | Firm FE | 1.2435 | | 0.0863 | 2.4007 | −1.0372 | 3.5242 | 0.2857 | 595 |
| Bloomberg | Time FE | −1.4881 | | −2.4582 | −0.5181 | −3.4000 | 0.4238 | 0.1277 | 595 |
| Bloomberg | Two-way FE | −0.1635 | | −1.2293 | 0.9022 | −2.2641 | 1.9370 | 0.8788 | 595 |
| LSEG | Pooled OLS | −0.1825 | ** | −0.2097 | −0.1553 | −0.2361 | −0.1289 | 0.0000 | 997 |
| LSEG | Firm FE | 0.0261 | | −0.0237 | 0.0760 | −0.0722 | 0.1245 | 0.6024 | 997 |
| LSEG | Time FE | −0.2377 | ** | −0.3242 | −0.1513 | −0.4082 | −0.0673 | 0.0064 | 997 |
| LSEG | Two-way FE | −0.1320 | | −0.2034 | −0.0607 | −0.2727 | 0.0086 | 0.0661 | 997 |

Table A.8.: Regression Results for Volatility (260-day; Standardized ESG Z-scores; 68% and 95% CI; Controls in ROCE and btm ratio)

| **Provider** | **Model** | $\beta$ | **Sig** | **CI68 Low** | **CI68 High** | **CI95 Low** | **CI95 High** | **p-value** | **N** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **Banks** | | | | | |
| Bloomberg | Pooled OLS | 0.0783 | | −0.2330 | 0.3895 | −0.5351 | 0.6916 | 0.8025 | 989 |
| Bloomberg | Firm FE | 0.0552 | | −0.7296 | 0.8401 | −1.4916 | 1.6020 | 0.9442 | 989 |
| Bloomberg | Time FE | −0.2099 | | −1.1246 | 0.7049 | −2.0128 | 1.5930 | 0.8196 | 989 |
| Bloomberg | Two-way FE | −1.2596 | | −2.4405 | −0.0787 | −3.5870 | 1.0677 | 0.2891 | 989 |
| LSEG | Pooled OLS | −1.9667 | ** | −2.3615 | −1.5718 | −2.7448 | −1.1885 | 0.0000 | 1474 |
| LSEG | Firm FE | −2.5402 | | −4.1985 | −0.8819 | −5.8085 | 0.7281 | 0.1279 | 1474 |
| LSEG | Time FE | −2.0239 | | −3.6208 | −0.4270 | −5.1712 | 1.1234 | 0.2077 | 1474 |
| LSEG | Two-way FE | −2.3050 | | −3.9969 | −0.6132 | −5.6394 | 1.0294 | 0.1757 | 1474 |
| | | | | **Energy** | | | | | |
| Bloomberg | Pooled OLS | −0.7018 | | −1.1083 | −0.2953 | −1.5029 | 0.0993 | 0.0860 | 595 |
| Bloomberg | Firm FE | 1.6734 | | 0.1161 | 3.2306 | −1.3958 | 4.7425 | 0.2857 | 595 |
| Bloomberg | Time FE | −2.0025 | | −3.3079 | −0.6971 | −4.5753 | 0.5703 | 0.1277 | 595 |
| Bloomberg | Two-way FE | −0.2201 | | −1.6543 | 1.2141 | −3.0467 | 2.6066 | 0.8788 | 595 |
| LSEG | Pooled OLS | −2.2580 | ** | −2.5945 | −1.9215 | −2.9213 | −1.5948 | 0.0000 | 997 |
| LSEG | Firm FE | 0.3235 | | −0.2938 | 0.9409 | −0.8932 | 1.5403 | 0.6024 | 997 |
| LSEG | Time FE | −2.9417 | ** | −4.0118 | −1.8716 | −5.0507 | −0.8327 | 0.0064 | 997 |
| LSEG | Two-way FE | −1.6337 | | −2.5169 | −0.7506 | −3.3743 | 0.1068 | 0.0661 | 997 |

# B. Appendix 2 – Implementation

This appendix provides a detailed description of the empirical implementation. All data preparation, regression analysis, and visualization were conducted using Python. The empirical framework was implemented using standard scientific libraries, including pandas for data handling, statsmodels and linearmodels for regression analysis, and matplotlib for graphical output.

The appendix outlines the data cleaning procedures, variable construction, panel data setup, and estimation methodology to ensure full transparency and reproducibility of the empirical results.

**Code Availability and Data Access** All code used for data preparation, regression analysis, and visualization is publicly available at: `https://github.com/Dragon999999999/ESG_Regression`.

Due to licensing restrictions, the underlying ESG and financial data obtained from Bloomberg and LSEG cannot be publicly shared. The repository therefore contains the full implementation code but does not include the proprietary datasets.

The expected data structure and variable definitions are documented in the repository to ensure transparency and reproducibility for researchers with authorized access to the respective data sources.